

30.3 A 24Gb 42.5Gb/s GDDR7 DRAM with Low-Power WCK Distribution, an RC-Optimized Dual-Emphasis TX, and Voltage/Time-Margin-Enhanced Power Reduction

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[Sang-Hoon Kim](#) ; [Jaehyeok Baek](#) ; [Moon-Chul Choi](#) ; [Daewoong Lee](#) ; [Donggun An](#) ; [Se mi Kim](#) ...

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As demand for high-resolution graphics processing, large-scale data analysis, and increasing data-processing requirements, driven by AI advancements, continues to surge, the importance of memory technologies, particularly high-performance GDDR, have grown significantly. PAM3 signaling was introduced in GDDR7 [1], [2] to achieve high data rates, which is essential for high-performance systems; resulting in significantly enhanced data processing speed. While high data rates are critical for fast data transfer, increasing memory density is equally important in providing the storage capacity needed for processing extensive datasets, thereby maximizing system performance. However, high bandwidth and density can lead to increased power consumption, and the elevated junction temperature. The high-performance system is required to maintain its temperature within a certain range for performance and reliability, as thermal throttling causes performance degradation.

This paper presents a 42.5Gb/s 24Gb GDDR7 DRAM fabricated in a 1b-DRAM process, achieving an energy efficiency improvement of 13.6%, compared to previous GDDR7 [1]. The circuit size shrank as process technology advanced, so an optimized floorplan is required. Figure 30.3.1 shows the floorplan including DQ placement, compared to prior work [1]. In this work, DQs are placed in two rows considering their power and speed, as shown in Fig. 30.3.1. The power consumption of high-speed clock trees increases dramatically with the increased operating frequency; moreover, clock-tree performance degradation occurs because increasing the driving strength does not improve the driving capability when the capacitance of clock trees is large enough compared to the operating frequency. Power consumption for clock distribution to DQ is reduced by



26%, compared to that of a 1-row DQ placement, due to the reduced longest-clock path, even though the number of repeaters for clock distribution is increased by the longer overall clock distribution length. Additionally, the 2-row architecture reduces the lengths of the longest data busses, resulting in fewer repeaters and lower power. The 2-row DQ placement also helps to minimize timing skew among data buses to DQs, due to ease of signal routing and repeater arrangement. On the other hand, the core access time of GDDR during write and read operations has been the shortest among other DRAMs due to the fastest speed. Therefore, the length of the global I/O (GIO) line, in the cell area, has been limited to support high-frequency operation. As shown in [Fig. 30.3.1](#), the array size, including the GIO length, is increased for high memory density. The more significant GIO loading causes the operating frequency to degrade and the power consumption to increase. Techniques to guarantee the core operation at high frequency and to optimize power consumption, even with a $1.5\times$ longer GIO compared to prior work, are employed and described in this paper.

The proposed WCK architecture, as shown in [Fig. 30.3.2](#), plays a crucial role in achieving energy efficiency. For prior GDDR7 [1], WCK operates in low- (LF) and high-frequency (HF) modes that use CMOS and CML circuitry. While CMOS offers reduced power consumption, compared to CML, it results in degraded clock characteristics, such as higher jitter, and reduced signal integrity. To decrease current consumption and minimize these clock-related degradations for frequencies beyond the CMOS circuits' operating limit frequency (f_{LF}), the proposed high-frequency low-power (HF_LP) mode is used. The HF_LP mode, like the LF mode, operates with CMOS circuits. To reduce variations in clock delay by supply drift, HF_LP mode employs an internally generated supply voltage (IVC) instead of relying on an external supply (EVC). An IVC of 0.85V is generated from a 1.2V EVC for CMOS circuits: the divider and repeaters for WCK distribution. Due to the CMOS IVC circuits, the current from VDDQ for WCK during both read operation including RCK toggling (I_{DD4R_RCK}) and write operation (I_{DD4W}) is reduced by 42% and 38%, compared to HF mode. The IVC is also adapted to the supply for CMOS circuits during LF mode, which operates near several GHz. Thus, the delay drift from WCK to DQ-out and DQ-in due to supply variation, t_{WCKDQ} , is reduced by 32% and 24%, compared to when EVC is applied during the operation at f_{LF} . A CMOS repeater is commonly used in LF and HF_LP mode, and the selected clock (WCK2_CMOS), according to the mode, is distributed to DQ. In HF mode, the clock for DQ (WCK2_CML) is distributed using a CML repeater. The clock suitable for the mode is selected, between WCK2_CML and WCK2_CMOS, and is used in the CML-to-CMOS (C2C) circuit for each DQ. The CK4 used for CA is also chosen between CML and CMOS_IVC clock, depending on the mode, like the WCK2 path.

[Figure 30.3.3](#) shows the implementation of transmitter (TX) and receiver (RX) circuits. The TX consists of a 4:1 multiplexer, an on-chip equalizer, segmented drivers for the main path, a feed-forward equalizer (FFE), and an AC equalizer (ACEQ) [2]. In prior work [1], the output impedance of the TX driver was implemented by a combination of two types of passive resistors, which are implemented with the ACT or BP layers. However, in this work, only the BP resistor is used to improve the linearity of the TX driver, by reducing the impact of variations. The BP resistor has a lower ($(1/10)$) sheet resistance than the ACT resistor, requiring a larger area to implement the required TX-driver output impedance. Thanks to the shared-R SST driver structure, not only is the level-mismatch ratio (RLM) and the bandwidth simultaneously improved [2], but the required resistor area is also reduced, making it possible to only use the BP resistor. In the TX, additional bandwidth enhancement has been achieved using both FFE and ACEQ. The DRAM metal-oxide-metal (MOM) capacitor, which is insensitive to capacitance variation due to voltage changes across its two terminals, is first implemented for ACEQ. A MOS capacitor is commonly used due to its large capacitance per unit area, but such capacitance varies with voltage difference between two terminals. To mitigate capacitance variations, a pair of bi-directional MOS capacitors is used. However, for one case, one of them is connected with its gate to the pad; the exposed gate, without a protection device, may be destroyed by an ESD event. Adding a CDM resistor to the TX output stage is necessary to prevent ESD breakdown, but it results in signal integrity (SI) degradation. Using the DRAM MOM capacitor, we implemented ACEQ without SI degradation, which is caused by the CDM resistor.

In the previous GDDR, the maximum density was implemented as 16Gb, and the corresponding GIO length allowed for the high-frequency operation as required. To support 24Gb density, the length of the GIO line increases up to $1.5\times$ compared to prior work as a one-bit address is added to the row addresses. Larger capacitances of this long GIO line result in a lower voltage margin, meaning performance degradation; thereby, reducing the maximum operating frequency and increasing power consumption. To improve the reduced voltage margin caused by the $1.5\times$ GIO line, a PMOS switch and cross-coupled PMOS were added to the local sense amplifier (LSA), as shown in [Fig. 30.3.4](#). These additional transistors allow the high level of LIO/LIOB to rise to V_{LSA} instead of V_{LIO} , increasing the amount of GIO developed. Power-reduction

techniques are also implemented for the core operation. As shown in [Fig. 30.3.4](#), GIO switches in the middle of GIO lines open when the near cell from the GIO driver is active to reduce parasitic loading, achieving 5.5% and 4.9% reduction in core power consumption for I_{DD4W} and I_{DD4R} each. The pulse width of the LSA enable signal (EN_LSA), determining the timing when LSA develops GIO/GIOB, is reduced in the near cell area. Also, the size of LSA_SRC transistors is split to reduce the LSA source current, especially in the near cell area. The techniques relevant to LSA operation allow the voltage margin between GIO and GIOB driven by LSA to remain similar regardless of the distance from the GIO driver or sense-amplifier, and the power consumption to be reduced by about 4.8% in I_{DD4R} .

To meet the high bandwidth requirements, the operating frequency increases, and the internal timing margin decreases. Meanwhile, supporting a wide V_{DD} range is important to obtain both a speed extension at a high-voltage supply and low-power consumption at a low-voltage supply. [Figure 30.3.5](#) describes the proposed on-chip EVC detector and its operation in covering a wide V_{DD} range. The on-chip EVC detector generates a 4b code that represents the current V_{DD} level and the strobe-signal delay, illustrated in Δt , is set according to the V_{DD} level; thereby, increasing the setup margin at a high-voltage level and the hold margin at a low-voltage level. This technique helps to achieve a maximum frequency operation, up to 42.5Gb/s at 1.2V and over 38.8Gb/s at 1.1V.

[Figure 30.3.6](#) shows the measurement results. The measured t_{CK} Shmoo plot shows the maximum data rate, 42.5Gb/s and 38.8Gb/s, at a 1.2 and 1.1V supply. When WCK operates in HF_LP mode, compared to HF mode, at the same operating frequency, lower than f_{HF} , I_{DDQ4W} and I_{DDQ4R_RCK} are reduced by 9.1% and 15.5%. Compared to previous work [\[1\]](#), [\[3\]](#), the power efficiency is improved by 36% over GDDR6 and 13.6% over previous GDDR7. A microphotograph of the implemented 24Gb GDDR7, fabricated in 1b DRAM technology and occupying 38.85 mm²/die, is shown in [Fig. 30.3.7](#).



Figure 30.3.1:
1-row and proposed 2-row DQ arrangement comparison.



Figure 30.3.2:
Proposed WCK architecture (top). Operational frequency-dependent WCK mode (bottom).



Figure 30.3.3:
PAM3 transmitter with FFE and ACEQ, and receiver (top). Comparison of SST drivers (bottom-left). Comparison of MOS and implemented DRAM-MOM capacitor (bottom-right).



Figure 30.3.4:
Structure of the local sense amplifier (LSA, left). Techniques for increasing the differential voltage (dGIO) between GIO and GIOB and reducing power consumption (right).



Figure 30.3.5:
Proposed on-chip EVC detector: Block diagram (top) and V_{DD} - t_{CK} Shmoo for the setup and hold margins between input and strobe signals (bottom).



Figure 30.3.6:

Measured results: $V_{DD} - t_{WCK}$ Shmoo (top), comparison of I_{DDQ4W}/I_{DDQ4W_RCK} for HF_LP and HF mode (bottom-left), and comparison of power efficiency (bottom-right).



Figure 30.3.7:

Chip micrograph and design summary table.